

[GLR17,CRYPTO] Message Franking via Committing Authenticated Encryption

SUMMARY: The authors formalize messaging franking's security goals, discuss existing AEAD schemes, and propose more efficient scheme to implement it.

1 Background

In 2017, Facebook proposed message franking that allows users to verifiably report a problematic message and its sender. At a high level, message franking cryptographically binds the sender's identity and the messages they sent in messaging, which enables the recipients to submit report with cryptographic verifiable proof.

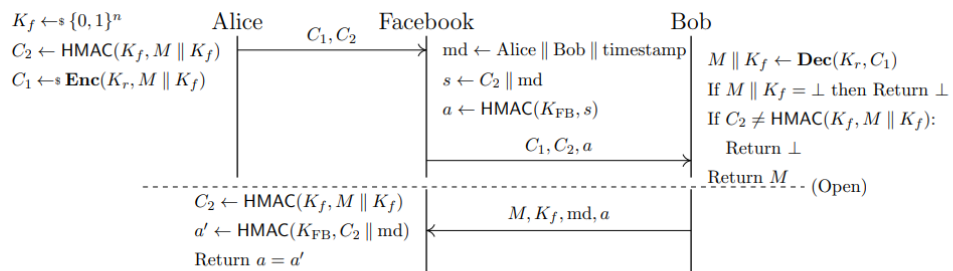


Figure 3: Facebook's message franking protocol [51]. The key K_r is a one-time-use symmetric key derived as part of the record layer protocol. The top portion is the sending of an encrypted message to the recipient. The bottom portion is the abuse reporting protocol.

In Figure 3, the use of HMAC avoids the platform storage for ciphertexts.

2 Preliminaries

The authors formalize the message franking schemes to a new cryptographic primitive: compactly committing authenticated encryption with associated data (ccAEAD). Before diving into the details of ccAEAD, we introduce the related cryptographic primitives as follows.

Committing encryption (CE) [GH03,preprint]

As its usage in many schemes, an encryption scheme can be viewed as a commitment to the encrypted message that takes the ciphertext and decryption key as the commitment and opening, respectively.

CE considers the security of an encryption and commitment at the same time. Unlike a conventional encryption, CE involves a verification algorithm in addition to (KG, Enc, Dec). Normally, it takes input of key, ciphertext, and message, verify the commitment via decrypting.

Tweakable block cipher [[LRW02](#), CRYPTO]

[symmetric - What is a tweakable block cipher? - Cryptography Stack Exchange](#)
xilinx.github.io/Vitis_Libraries/security/2020.1/guide_L1/internals/xts.html

Tweakable block cipher (e.g., three fish) introduces a tweak in addition to the key. Both key and tweak are used to select a particular permutation from the pseudopermutation family. Moreover, the additional input resolves the problem of related key attack. Furthermore, it is cheap to change a tweak, but expensive to change a key (in a tweakable cipher).

Here, what confuse me is the difference between tweakable block cipher and nonce-based encryption (Here, the former also indicates the mode of encryption such as XTS). Intuitively, nonce-based is contained in tweakable, which takes the nonces as tweak. However, nonce-based is more simple and maintains the same security level as the tweakable cipher.

Another point that confuse me is why changing key is expensive in AES? From the answers in StackExchange, I guess it is because that some intermediate results can be reused to encrypt different messages, when the key is fixed. Therefore, if the tweak is only related to the message, it will be cheap to change.

Nonce-based encryption [[Rogaway04](#), FSE]

Instead of using random initial value (IV) to encrypt a message, nonce-based encryption consider the case that IV is a nonce (number use only once). With this assumption, the security of encryption can be significantly clarified and improved.

This idea inspires the design of many cryptographic schemes. For example, $E_k(E_k(N) + i)$ is a nonce-based PRG, where E is a block cipher, N is a nonce, and i is a counter that records the order of blocks.

Authenticated encryption with associate data (AEAD)

A private-key encryption scheme is an authenticated encryption (AE) scheme if it is CCA-secure and unforgeable. -- IMF 3rd edition

Associate data such as header information usually requires integrity; therefore, it is a waste to directly encrypts it with AE. Hence, some schemes are designed to solve this problem in efficient.

3 Design Goals

Security

- **Confidentiality & Integrity:** These two properties are similar to their definitions in conventional AEAD except the adoption to the opening security.
- **Sender binding:** The sender's message is bound to the message it actually sent, which prevents evading reporting.
- **Receiver binding:** This is similar to binding in conventional commitment. A malicious receiver cannot open the commitment to a message that differs from the actual one, which prevents malicious reporting.
- **Multiple opening (MO) security:** Opening a commitment with the encryption key may compromise the security of ciphertexts that were encrypted by the same key. Therefore, multiple opening security is required in this case. Note that, in Signal (i.e., the double ratchet algorithm), each key is used only once, and single opening is sufficient.

Performance

- **Compactly committing (cc):** This means the commitment part of the ciphertext is small, and is linear in the key length instead of the message length. In contrast, conventional CE always takes the entire ciphertext as commitment.

Discussion

- Full robustness implies receiver binding, and the converse is not true. Because full robustness does not restrict the challenge message; it only requires the adversary to find a distinct key that decrypts the ciphertext to a non-empty plaintext.

$\text{r-BIND}_{\text{SE}}^A:$ $((H, M, K), (H', M', K'), C) \leftarrow_s \mathcal{A}$ $\bar{M} \leftarrow \text{dec}(K, H, C)$ $\bar{M}' \leftarrow \text{dec}(K', H', C)$ $\text{Return } (M = \bar{M}) \wedge (M' = \bar{M}') \wedge ((H, M) \neq (H', M'))$	$\text{FROB}_{\text{SE}}^A:$ $((H, K), (H', K'), C) \leftarrow_s \mathcal{A}$ $\text{If } K = K' \text{ then Return false}$ $M \leftarrow \text{Dec}(K, H, C)$ $M' \leftarrow \text{Dec}(K', H', C)$ $\text{Return } (M \neq \perp) \wedge (M' \neq \perp)$
---	---

Figure 17: **(Left)** An equivalent formulation of receiver binding for the case of traditionally committing encryption schemes. **(Right)** The full robustness security game.

4 Existing Scheme

The authors first consider whether existing AEAD schemes satisfy committing AEAD's security goals, and then propose a new scheme of ccAEAD (i.e., CEP). The below figure summarizes the security and performance of these schemes.

Scheme	MO security	Sender binding	Rec. binding	Enc	Dec	Ver
Encode-then-Encipher (Ideal)		✓	✓	–	–	–
Encrypt-then-HMAC (one key)		✓	✓	2+1	2+1	2+1
HMAC-then-CBC		✓	✓	2+1	2+1	2+1
CtE1	✓	✓	✓	3+1	3+1	1+1
CtE2 (Facebook)	✓	✓	✓	3+2	3+2	1+1
CEP	✓	✓	✓	2+1	2+1	1+1

Figure 1: Summary of schemes investigated in this work. The columns indicate whether the scheme meets multiple-opening (MO) security, sender binding, and receiver binding. The last three columns indicate the number of cryptographic passes over a bit string of length equal to the message plus the number of passes needed to handle the associated data, for each of the three main operations. We omit comparisons with concrete encode-then-encipher constructions, which vary in the number of passes required.

Committing Encode-then-Encipher (EtE) [BR00,ASIACRYPT]

$\text{Enc}(K, H, M):$ $R \leftarrow \{0, 1\}^r$ $C \leftarrow \tilde{E}_K^H(M \parallel R \parallel 0^s)$ $\ell \leftarrow l + r + s - t$ $C_1 \leftarrow C[1, \dots, \ell]$ $C_2 \leftarrow C[\ell + 1, \dots, l + r + s]$ Return (C_1, C_2)	$\text{Dec}(K, H, C_1, C_2):$ $M' \parallel R' \parallel Z \leftarrow \tilde{D}_K^H(C_1 \parallel C_2)$ If $Z \neq 0^s$ then Return $\perp$ Return $(M', (R', K))$	$\text{Ver}(H, M, K_f, C_2):$ $R \parallel K \leftarrow K_f$ $\ell \leftarrow l + r + s - t$ $C \leftarrow \tilde{E}_K^H(M \parallel R \parallel 0^s)$ Return $C[\ell + 1, \dots, l + r + s] = C_2$
--	--	---

Figure 6: Encode-then-encipher as a committing AEAD scheme where the commitment is the final t bits of the ciphertext. $\tilde{E}_K^H$ and $\tilde{D}_K^H$ refer to encryption and decryption for a tweakable blockcipher TC where the header H is the tweak and K is the key.

Encrypt-then-MAC (EtM)

$\text{Enc}(K, H, M):$ $K^e \leftarrow \text{KDF}_K(0)$ $K^m \leftarrow \text{KDF}_K(1)$ $R \leftarrow \mathcal{R}$ $R \parallel C \leftarrow \text{enc}_{K^e}(M; R)$ $T \leftarrow F_{K^m}(H \parallel R \parallel C)$ Return $(R \parallel C, T)$	$\text{Dec}(K, H, C_1, C_2):$ $R \parallel C \leftarrow C_1$ $K^e \leftarrow \text{KDF}_K(0)$ $K^m \leftarrow \text{KDF}_K(1)$ $T' \leftarrow F_{K^m}(H \parallel R \parallel C_1)$ If $T' \neq C_2$ then Return $\perp$ $M \leftarrow \text{dec}_{K^e}(C_1)$ If $M = \perp$ then Return $\perp$ Return $(M, (R, K))$	$\text{Ver}(H, M, (R, K), C_2):$ $K^e \leftarrow \text{KDF}_K(0)$ $K^m \leftarrow \text{KDF}_K(1)$ $C \leftarrow \text{enc}_{K^e}(M; R)$ $T \leftarrow F_{K^m}(H \parallel R \parallel C)$ Return $T = C_2$
---	---	--

Figure 8: Committing AEAD scheme EtM[KDF, F , SE] that composes an encryption scheme SE = (kg, enc, dec) using random coins from $\mathcal{R}$, a MAC F , and that derives keys via a function KDF.

- When the enc. key and PRF key is chosen independently, this scheme is not receiver binding since the receiver can submit a different but valid key K_e in reporting. Two (H, K) pairs can map to the same (C, R) .
- Q: The above problem can resolved by assume the Encryption is full robustness?

MAC-then-Encrypt (MtE)

$\text{Enc}(K, H, M):$ $K^e, K^m \leftarrow K$ $IV \leftarrow \{0, 1\}^n$ $T \leftarrow \text{RO}_{K^m}(H \parallel M)$ $C \leftarrow \text{CBC}_{K^e}(\text{Pad}_n(M \parallel T); IV)$ $\ell \leftarrow \text{Pad}_n(M \parallel T)/n$ $C' \parallel C_{\ell-2} \parallel C_{\ell-1} \parallel C_\ell \leftarrow C$ $\text{Return } (C', IV \parallel C_{\ell-2} \parallel C_{\ell-1} \parallel C_\ell)$	$\text{Dec}(K, H, C_1, C_2):$ $K^e, K^m \leftarrow K$ $IV \parallel C_{\ell-2} \parallel C_{\ell-1} \parallel C_\ell \leftarrow C_2$ $C_f \leftarrow C_{\ell-2} \parallel C_{\ell-1} \parallel C_\ell$ $M \parallel T \leftarrow \text{CBC}_{K^e}^{-1}(C_1 \parallel C_f; IV)$ $T' \leftarrow \text{RO}_{K^m}(H \parallel M)$ $\text{If } T \neq T' \text{ then Return } \perp$ $\text{Return } (M, (K^e, K^m))$	$\text{Ver}(H, M, K_f, C_2):$ $K^e, K^m \leftarrow K_f$ $IV \parallel C'_{\ell-2} \parallel C'_{\ell-1} \parallel C'_\ell \leftarrow C_2$ $T \leftarrow \text{RO}_{K^m}(H \parallel M)$ $P \leftarrow \text{Pad}_n(M \parallel T)$ $C \leftarrow \text{CBC}_{K^e}(P; IV)$ $C' \parallel C''_{\ell-2} \parallel C''_{\ell-1} \parallel C''_\ell \leftarrow C$ $\text{Return } \bigwedge_{i=\ell-2}^{\ell} (C''_i = C'_i)$
--	--	--

Figure 10: Committing authenticated encryption based on MtE composition of CBC mode and a MAC modeled as a random oracle. The length ℓ is defined to be $\text{Pad}_n(M \parallel T)/n$. The function Pad is the standard (min-length) PKCS#7 padding used in TLS. The notation $\text{CBC}_K(\cdot; IV)$ and $\text{CBC}_K^{-1}(\cdot; IV)$ means CBC mode encryption and decryption with key K and initialization vector IV .

- This scheme is secure even the two keys are chosen independently.

Commitment-then-Encrypt (CtE, this formalizes Facebook's scheme)

$\text{CtE1-Enc}(K, H, M)$ $(K_f, C_2) \leftarrow \text{Com}(H \parallel M)$ $C_1 \leftarrow \text{enc}_K(C_2, M \parallel K_f)$ $\text{Return } (C_1, C_2)$	$\text{CtE1-Dec}(K, H, C_1, C_2)$ $(M \parallel K_f) \leftarrow \text{dec}_K(C_2, C_1)$ $\text{If } M = \perp \text{ then Return } \perp$ $b \leftarrow \text{VerC}(K_f, C_2, H \parallel M)$ $\text{If } b = 0 \text{ then}$ $\quad \text{Return } \perp$ $\text{Return } (M, K_f)$	$\text{CtE2-Dec}(K, H, C_1, C_2)$ $(M \parallel K_f) \leftarrow \text{dec}_K(H, C_1)$ $\text{If } M = \perp \text{ then Return } \perp$ $b \leftarrow \text{VerC}(K_f, C_2, H \parallel M)$ $\text{If } b = 0 \text{ then}$ $\quad \text{Return } \perp$ $\text{Return } (M, K_f)$
--	--	--

Figure 12: Algorithms for two Commit-then-Encrypt variants. Facebook's scheme uses CtE2 with an HMAC-based commitment. CtE1-Ver and CtE2-Ver both just output $\text{VerC}(H, M, K_f, C_2)$.

- CtE1 is better than CtE2
 - Commitment should be unique commitment in CtE2. For any (H,M) pair, only one valid commitment exists
 - Associated data only cryptographic processed once

5 Proposed Scheme (Committing Encrypt-and-PRF, CEP)

$\text{CEP-Enc}_K^N(H, M):$ $m \leftarrow \lceil M /n \rceil$ $P \leftarrow G(K, N, m + 2n)$ $C_1 \leftarrow (P_2 \parallel \dots \parallel P_{m+1}) \oplus M$ $C_2 \leftarrow F_{P_0}^{\text{cr}}(H \parallel M)$ $T \leftarrow F_{P_1}(C_2)$ Return $(C_1 \parallel T, C_2)$	$\text{CEP-Dec}_K^N(H, C_1 \parallel T, C_2):$ $m \leftarrow \lceil C_1 /n \rceil$ $P \leftarrow G(K, N, m + 2n)$ $M \leftarrow (P_2 \parallel \dots \parallel P_{m+1}) \oplus C_1$ $C'_2 \leftarrow F_{P_0}^{\text{cr}}(H \parallel M)$ $T' \leftarrow F_{P_1}(C'_2)$ If $T \neq T' \vee C'_2 \neq C_2$ then Return $\perp$ Return (M, P_0)	$\text{CEP-Ver}(H, M, K_f, C_2):$ $C'_2 \leftarrow F_{K_f}^{\text{cr}}(H \parallel M)$ If $C'_2 \neq C_2$ then Return 0 Return 1
---	---	---

Figure 15: The nonce-based committing AEAD scheme $\text{CEP}[G, F, F^{\text{cr}}]$. For the pad P generated by G , the notation P_i refers to the i th n -bit block of P , i.e. $P[in, \dots, (i+1)n-1]$.

Tips

- Tag T preserves ciphertext integrity.
- P_0 will be public in commitment verification. (H, M, K_f, P_0, C_2) is the opening.

Pros

- Nonce-based advantages
- Ciphertext expansion is reduced
- encryption and decryption process less input
- Achieves multiple-opening security goal